

Fake USB / SSD Investigation

Overview

This repository documents the investigation of a counterfeit external USB storage device purchased online.

The device claimed to provide 16TB of storage but exhibited unusual behaviour during filesystem operations and capacity testing.

The investigation was performed using Linux command-line utilities, capacity validation tools, and a physical teardown of the enclosure.

Symptoms Observed

Symptoms

- Filesystem instability
- Partitioning failures
- Storage corruption
- Inconsistent device behaviour
- Capacity reporting mismatches
- Unusual controller identification

Investigation Workflow

Investigation Workflow

The following Linux tools were used during validation:

- fdisk
- lsusb
- f3probe
- smartctl
- dmesg

Capacity Verification

Capacity Verification

The device initially reported itself as a 64GB flash device.

However, testing with `f3probe` revealed the true usable capacity was approximately 11GB.

This indicates manipulated firmware or a spoofed flash controller designed to falsely advertise storage capacity.

Important output:

Usable size: 11.01 GB
Announced size: 62.50 GB

```
(michael@vengeance)-[~]  
$ sudo fdisk -l /dev/sda  
Disk /dev/sda: 62.5 GiB, 67108864000 bytes, 131072000 sectors  
Disk model: Flash Reader  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
(michael@vengeance)-[~]
$ sudo f3probe --destructive --time-ops /dev/sda
[sudo] password for michael:
F3 probe 9.0
Copyright (C) 2010 Digirati Internet LTDA.
This is free software; see the source for copying conditions.

WARNING: Probing normally takes from a few seconds to 15 minutes, but
         it can take longer. Please be patient.

Bad news: The device `/dev/sda' is a counterfeit of type limbo

You can "fix" this device using the following command:
f3fix --last-sec=23084031 /dev/sda

Device geometry:
    *Usable* size: 11.01 GB (23084032 blocks)
    Announced size: 62.50 GB (131072000 blocks)
    Module: 64.00 GB (2^36 Bytes)
    Approximate cache size: 0.00 Byte (0 blocks), need-reset=no
    Physical block size: 512.00 Byte (2^9 Bytes)

Probe time: 259.5ms
Operation: total time / count = avg time
    Read: 111.2ms / 2154 = 51us
    Write: 146.7ms / 2150 = 68us
    Reset: 0us / 0 = 0us
```

USB Controller Identification

```
## USB Controller Identification
```

Using `lsusb`, the device identified itself as:

Chipsbank Microelectronics Co., Ltd Flash Disk

Counterfeit storage devices commonly rely on low-cost programmable flash controllers capable of spoofing storage capacity values.

```
(michael@vengeance)-[~]  
$ lsusb  
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub  
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub  
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub  
Bus 003 Device 004: ID 0bda:0129 Realtek Semiconductor Corp. RTS5129 Card Reader Controller  
Bus 003 Device 005: ID 04f2:b802 Chicony Electronics Co., Ltd HD Webcam  
Bus 003 Device 008: ID 093a:2510 Pixart Imaging, Inc. Optical Mouse  
Bus 003 Device 038: ID 8087:0033 Intel Corp. AX211 Bluetooth  
Bus 003 Device 044: ID 1e3d:198a Chipsbank Microelectronics Co., Ltd Flash Disk  
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
```

Physical Teardown

This becomes the strongest section now.

Physical Teardown

The enclosure was carefully opened for physical inspection.

Observations:

- No legitimate NVMe SSD present
- No visible NAND flash packages
- No standard SSD controller architecture
- Large hollow cavity inside enclosure
- Excess adhesive/glue covering PCB area
- Minimal USB interface hardware only



Conclusion

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The investigation confirmed the device was counterfeit.

Evidence included:

- spoofed storage reporting
- failed filesystem behaviour
- false advertised capacity
- abnormal internal hardware design
- absence of legitimate SSD components

This repository demonstrates a basic Linux-based workflow for identifying counterfeit storage devices using both software validation and physical inspection.